

Small-Sample Corrections for Likelihood Ratio Tests in Structural Equation Modeling

A Monte Carlo Comparison

Master's Thesis

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Chapter 1

Introduction

1.1 Background and Motivation

Structural equation modeling (SEM) has established itself as one of the most versatile and widely used statistical frameworks in the social, behavioral, and health sciences. Its capacity to model complex relationships among latent variables, account for measurement error, and test theoretically derived hypotheses simultaneously has made it an indispensable tool for modern empirical research. Applications of SEM range from confirmatory factor analysis of psychological instruments to complex mediation and moderation models in education, marketing, and epidemiology. The flexibility of SEM—encompassing path analysis, latent growth curve modeling, and multi-group comparisons—ensures its continued prominence across disciplines.

At the core of SEM inference lies the likelihood ratio test (LRT), commonly referred to as the model chi-square test. The standard LRT statistic, denoted T_{ML} , is computed as $(n - 1)F_{\text{ML}}$, where F_{ML} is the maximum likelihood discrepancy function and n is the sample size. Under the null hypothesis that the population covariance matrix is perfectly reproduced by the model, T_{ML} is asymptotically distributed as a central chi-square random variable with degrees of freedom equal to the difference between the number of unique elements in the sample covariance matrix and the number of freely estimated parameters. This result, rooted in Wilks' (1938) theorem, provides the theoretical justification for the ubiquitous chi-square test in SEM.

Despite its theoretical elegance, the chi-square reference distribution is only a first-order asymptotic approximation. The approximation relies on the sample size being sufficiently large relative to the number of model parameters—a condition that is frequently violated in practice. Many SEM applications in psychology, education, and health sciences involve sample sizes below 200, and sometimes as small as 30 to 50 observations. In such situations, the chi-square approximation can be severely inadequate, leading to inflated Type I error rates, rejection of correctly specified models, and misleading conclusions

about model fit.

1.2 The Small-Sample Problem

The small-sample behavior of the LRT in SEM has been a persistent concern since the earliest days of the methodology. The fundamental issue is that the expected value of T_{ML} under the null hypothesis deviates from its nominal degrees of freedom when the sample size is finite. Specifically, $\mathbb{E}[T_{\text{ML}}] = df + O(n^{-1})$, meaning that the test statistic is systematically inflated in small samples. This inflation translates directly into elevated Type I error rates: the test rejects the null hypothesis more frequently than the nominal significance level α would suggest.

Bartlett (1937) was among the first to address this problem in a general likelihood ratio context, proposing a multiplicative correction that adjusts the test statistic so that its expected value more closely matches the reference distribution. Swain (1975) later derived a closed-form version of the Bartlett correction specifically for covariance structure models, expressing the correction factor as a simple function of sample size n , the number of observed variables p , and the number of free parameters t . The Swain correction reduces the approximation error from $O(n^{-1})$ to $O(n^{-2})$, offering substantial improvements in small-sample inference.

Subsequent developments expanded the toolkit of small-sample corrections. Satorra and Bentler (1988, 1994) introduced a scaling correction for situations where the assumption of multivariate normality is violated, using a sandwich estimator of the asymptotic covariance matrix to adjust the chi-square statistic. Their mean- and variance-adjusted statistics, T_{SB} and $T_{\text{SB-MV}}$, have become standard outputs in SEM software. Yuan and Bentler (1997, 1998) proposed further refinements that combine scaling and mean-adjustment corrections, providing improved control under both normal and non-normal conditions. Browne (1982) and Shapiro (1984) developed an F -corrected version of the LRT that substitutes the chi-square reference distribution with an F -distribution, offering an alternative approach to small-sample adjustment. Bollen and Stine (1992, 1993) introduced a bootstrap-based correction that transforms the data to satisfy the null hypothesis exactly before resampling, providing a non-parametric approach that makes minimal distributional assumptions.

Despite the availability of these seven correction methods—the standard LRT, Satorra–Bentler scaled statistic, Yuan–Bentler T_2 , Bartlett correction, Swain correction, F -corrected LRT, and Bollen–Stine bootstrap—there is no comprehensive Monte Carlo comparison that evaluates their relative performance across a range of sample sizes and model types under normal data conditions. Existing studies typically compare only a subset of these methods or focus on a single model type, leaving practitioners without clear guidance on which correction to prefer in specific situations. This thesis addresses that gap by

conducting a systematic Monte Carlo simulation that evaluates all seven methods across three population models and twelve sample sizes.

1.3 Research Questions

The present study is guided by three overarching research questions that address the performance, robustness, and convergence behavior of the seven correction methods.

RQ1: How do different correction methods control Type I error rates across sample sizes and model types? This question investigates the empirical rejection rates of each corrected test at the nominal $\alpha = 0.05$ level. By examining rejection rates across twelve sample sizes ranging from $N = 20$ to $N = 1000$ and three models of varying complexity, we aim to characterize the sample-size–method interactions that determine test performance.

RQ2: Which methods achieve robust Type I error rates according to Bradley’s (1978) criterion? Bradley’s liberal criterion defines a test as robust if the empirical Type I error rate falls within the interval $[0.9\alpha, 1.1\alpha]$ for a nominal level α . At $\alpha = 0.05$, this corresponds to the interval $[0.045, 0.055]$. We classify each method–model–sample-size combination as robust, liberal, or conservative based on this framework, providing actionable guidance for practitioners.

RQ3: At what sample size does each corrected test converge to the nominal level? Convergence is defined as the minimum sample size at which the empirical Type I error rate first enters Bradley’s robust interval and remains within it for all larger sample sizes. This threshold identifies the minimum sample size required for reliable inference with each correction method, informing study design decisions.

1.4 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 provides a comprehensive literature review covering the theoretical foundations and mathematical formulations of all seven correction methods, as well as Bradley’s (1978) robustness criterion for evaluating Type I error rates. Chapter 3 describes the Monte Carlo simulation methodology in detail, including the population models, factorial design, implementation in R, and evaluation criteria. Chapter 4 presents the simulation results, including convergence rates, Type I error rates by model and sample size, mean absolute deviations from the nominal level, and cross-method comparisons. Chapter 5 discusses the implications of the findings, provides practical recommendations, acknowledges limitations, and suggests directions for future research. Chapter 6 describes the computational implementation, including the R package architecture, simulation flow, seed strategy, and fault isolation mechanisms. Chapter 7 concludes with a summary of contributions and closing remarks. Two appendices provide complete model specifications and representative R code for the simulation framework.

Chapter 2

Literature Review

This chapter provides a detailed review of the statistical theory underlying the likelihood ratio test in structural equation modeling and the seven small-sample correction methods examined in this thesis. For each method, we present the mathematical formulation, underlying assumptions, key references, and practical considerations for implementation. The chapter concludes with a discussion of Bradley's (1978) robustness criterion, which serves as the primary evaluation framework for the Monte Carlo simulation.

2.1 The Standard Likelihood Ratio Test

2.1.1 Mathematical Formulation

The maximum likelihood fitting function for SEM under multivariate normality is defined as:

$$F_{\text{ML}}(\boldsymbol{\theta}) = \ln |\boldsymbol{\Sigma}(\boldsymbol{\theta})| + \text{tr} [\mathbf{S} \boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}] - \ln |\mathbf{S}| - p \quad (2.1)$$

where $\boldsymbol{\Sigma}(\boldsymbol{\theta})$ is the model-implied covariance matrix as a function of the parameter vector $\boldsymbol{\theta}$, $\mathbf{S}$ is the sample covariance matrix, and p is the number of observed variables. The function F_{ML} is non-negative and equals zero if and only if $\boldsymbol{\Sigma}(\hat{\boldsymbol{\theta}}) = \mathbf{S}$, representing a perfect fit.

The standard LRT chi-square statistic is computed as:

$$T_{\text{ML}} = (n - 1) F_{\text{ML}}(\hat{\boldsymbol{\theta}}) \quad (2.2)$$

where n is the sample size and $\hat{\boldsymbol{\theta}}$ is the maximum likelihood estimate of the model parameters. Under the null hypothesis that the specified model holds exactly in the population, Wilks' (1938) theorem guarantees that:

$$T_{\text{ML}} \xrightarrow{d} \chi_{df}^2 \quad \text{as } n \rightarrow \infty \quad (2.3)$$

where $df = p(p + 1)/2 - t$ is the degrees of freedom, computed as the difference between the number of unique elements in the sample covariance matrix and the number of freely estimated parameters t .

2.1.2 Asymptotic Limitations

The chi-square reference distribution in Equation (2.3) is a first-order asymptotic result. Specifically, the cumulative distribution function satisfies:

$$P(T_{\text{ML}} \leq x) = P(\chi_{df}^2 \leq x) + O(n^{-1}) \quad (2.4)$$

This $O(n^{-1})$ error term becomes non-negligible when the ratio n/t is small. In practice, the expected value of T_{ML} exceeds the nominal degrees of freedom by an amount that depends on n , p , and t , leading to inflated rejection rates. The magnitude of this inflation increases as the model complexity grows relative to the available sample size.

Wilks (1938) acknowledged these limitations in his original paper, noting that the approximation relies on regularity conditions that may not hold in finite samples. Subsequent work by Bartlett (1937), Lawley (1956), and others sought to quantify and correct the finite-sample bias, giving rise to the correction methods reviewed in the following sections.

2.2 Satorra–Bentler Scaled Chi-Square (T_{SB})

2.2.1 Mathematical Formulation

The Satorra–Bentler scaled chi-square statistic adjusts the standard LRT for violations of multivariate normality by incorporating a scaling factor derived from the fourth-order moments of the observed variables. The corrected statistic is:

$$T_{\text{SB}} = \frac{T_{\text{ML}}}{\hat{c}} \quad (2.5)$$

where the scaling correction factor $\hat{c}$ is computed as:

$$\hat{c} = \frac{\text{tr} \left[(\hat{\mathbf{\Gamma}} \hat{\mathbf{W}})^2 \right]}{\text{tr} \left[\hat{\mathbf{\Gamma}} \hat{\mathbf{W}} \right]^2 / df} \quad (2.6)$$

In this expression, $\hat{\mathbf{\Gamma}}$ is an estimate of the asymptotic covariance matrix of the sample covariance vector under non-normality (incorporating kurtosis information), and $\hat{\mathbf{W}}$ is the estimated weight matrix. The scaling factor $\hat{c}$ equals unity under multivariate normality and exceeds unity when the data exhibit excess kurtosis, reflecting the additional sampling

variability introduced by non-normality.

An alternative and more compact expression for the scaling factor uses the trace of the product of the model-based and robust weight matrices:

$$\hat{c} = \frac{\text{tr}(\hat{\mathbf{\Gamma}})}{df} \quad (2.7)$$

Under the null hypothesis, T_{SB} is approximately distributed as χ^2_{df} , with the scaling correction recovering the correct Type I error rate when the normality assumption is violated.

2.2.2 Assumptions and Properties

The Satorra–Bentler correction assumes that the fourth-order moments of the observed variables exist and are finite. It is designed specifically for continuous observed variables and may not perform optimally with ordinal or severely non-normal data. The correction addresses the variance of the test statistic (second-moment correction) but does not adjust the mean, which can result in residual bias in very small samples. Satorra and Bentler (2001) extended the scaling correction to difference tests between nested models, which is important for comparing alternative model specifications.

Under multivariate normality, the scaling factor $\hat{c} \approx 1$ and $T_{\text{SB}} \approx T_{\text{ML}}$, so the correction has negligible effect. The primary benefit of the Satorra–Bentler correction manifests under conditions of moderate to severe non-normality, where the uncorrected T_{ML} can exhibit substantial Type I error inflation.

2.2.3 Implementation

In R, the Satorra–Bentler scaled statistic is computed automatically by the `lavaan` package (Rosseel, 2012) when the estimator is specified as `MLR` (maximum likelihood with robust standard errors). The function `scaledTestStatistic()` extracts the scaling factor directly. For models fitted with the standard `ML` estimator, the scaling factor can be computed post hoc using the robust weight matrix available through `lavInspect(fit, "gamma")`.

2.3 Yuan–Bentler T_2 Statistic

2.3.1 Mathematical Formulation

Yuan and Bentler (1997, 1998) proposed a mean- and variance-adjusted test statistic that combines elements of both the Satorra–Bentler scaling correction and the Bartlett

correction framework. The T_2 statistic is defined as:

$$T_{\text{YB}} = \frac{T_{\text{ML}}}{\bar{c}} \quad (2.8)$$

where $\bar{c}$ is a mean scaling correction factor that adjusts both the expected value and the variance of the test statistic. Unlike the Satorra–Bentler correction, which uses only a single scaling factor, the Yuan–Bentler approach computes:

$$\bar{c} = \frac{\hat{a}}{\hat{a}^2 + \hat{b}} \quad (2.9)$$

where $\hat{a}$ and $\hat{b}$ are derived from the traces involving the asymptotic covariance matrices. The corrected statistic T_{YB} is then compared to a scaled chi-square distribution $\bar{c}^{-1} \chi_{df}^2$, which adjusts both the mean and the variance of the reference distribution.

2.3.2 Assumptions and Properties

The Yuan–Bentler T_2 statistic is designed to perform well under both normal and non-normal conditions, making it a more general-purpose correction than the Satorra–Bentler scaled statistic. It accounts for kurtosis through the same fourth-moment structure used in the Satorra–Bentler correction, but the mean-adjustment component provides additional small-sample protection. Yuan and Bentler (1998) showed through simulation that T_2 maintains better Type I error control than T_{SB} in very small samples ($N < 100$) under moderate non-normality.

The correction assumes continuous observed variables and finite fourth-order moments. Computational stability can be an issue when the mean correction factor $\bar{c}$ is close to zero or negative, which may occur in very small samples with complex models. In such cases, the statistic defaults to the standard T_{ML} or the Satorra–Bentler correction, depending on the software implementation.

2.3.3 Implementation

In `lavaan`, the Yuan–Bentler statistic is available through the MLM estimator, which computes both mean- and variance-adjusted chi-square statistics. The `scaledTestStatistic()` function with the argument `mean.structure = TRUE` retrieves the mean correction factor and the adjusted statistic. The `semTools` package provides additional convenience functions for extracting and comparing corrected test statistics across multiple models.

2.4 Bartlett Correction (T_{Bart})

2.4.1 Mathematical Formulation

The Bartlett correction adjusts the LRT statistic by a multiplicative factor chosen so that the expected value of the corrected statistic matches the nominal degrees of freedom. The standard Bartlett correction for covariance structure models is:

$$c = 1 - \frac{2t + 2p + 3}{2n - 2t - 2p - 1} \quad (2.10)$$

where n is the sample size, t is the number of free parameters, and p is the number of observed variables. The corrected test statistic is:

$$T_{\text{Bart}} = c \cdot T_{\text{ML}} \quad (2.11)$$

which is referenced to the χ^2_{df} distribution.

An alternative, more numerically stable formulation of the correction factor is:

$$c = \frac{n - (2p + 5)/6 - (2t + p)/3}{n} \quad (2.12)$$

This alternative form ensures that $0 < c < 1$ for all valid combinations of n , p , and t , whereas the standard formula in Equation (2.10) can yield negative values when the denominator becomes negative (i.e., when $2n < 2t + 2p + 1$). For sample sizes below $n = 100$, the alternative formula is recommended.

2.4.2 Theoretical Properties

The Bartlett correction achieves second-order accuracy in the sense that:

$$P(T_{\text{Bart}} \leq x) = P(\chi^2_{df} \leq x) + O(n^{-2}) \quad (2.13)$$

This represents a substantial improvement over the uncorrected test, which has an $O(n^{-1})$ approximation error. The correction works by matching the first moment of the test statistic to the reference distribution: $\mathbb{E}[T_{\text{Bart}}] = df + O(n^{-2})$ rather than $\mathbb{E}[T_{\text{ML}}] = df + O(n^{-1})$.

The derivation of the Bartlett correction for SEM relies on the cumulant expansion of the log-likelihood function. Under multivariate normality, the third-order cumulants of the log-likelihood derivatives vanish due to the symmetry of the normal distribution, and the fourth-order cumulants take a structured form that depends only on the model geometry. Lawley (1956) provided the general formula for computing the Bartlett correction from these cumulants, and Swain (1975) applied Lawley's formula to covariance structure

models to obtain the closed-form expressions in Equations (2.10) and (2.12).

2.4.3 Implementation

The Bartlett correction is computed in R using the `chisqSmallN()` function from the `semTools` package, which implements both the standard and alternative formulas, automatically selecting the appropriate version based on the sample size. The correction can also be computed manually using the formulas above once n , p , t , and df are known. In this study, we use the wrapper function `safe_chisqSmallN()`, which handles edge cases such as negative correction factors and returns a logical rejection decision along with the corrected statistic.

2.5 Swain's Correction (T_{Swain})

2.5.1 Mathematical Formulation

Swain (1975) derived the Bartlett correction specifically for the class of covariance structure models used in SEM. Although the Swain correction is mathematically equivalent to the Bartlett correction described in Section 2.4, it is often treated as a distinct method in the SEM literature because Swain's derivation was the first to express the correction in terms of the structural parameters specific to covariance structure models.

The Swain correction factor is:

$$c_{\text{Swain}} = 1 - \frac{2t + 2p + 3}{2n - 2t - 2p - 1} \quad (2.14)$$

and the corrected statistic is:

$$T_{\text{Swain}} = c_{\text{Swain}} \cdot T_{\text{ML}} \quad (2.15)$$

Swain's contribution was to show that the general Lawley (1956) formula reduces to this simple closed-form expression for covariance structure models under multivariate normality. The key insight is that the complex cumulant calculations simplify dramatically when the model is expressed in terms of covariance matrices rather than raw parameter vectors, because the Wishart distribution imposes a regular structure on the fourth-order cumulants.

2.5.2 Practical Considerations

In practice, the Swain and Bartlett corrections produce identical results when applied to the same model. However, the alternative stable formulation in Equation (2.12) is sometimes attributed specifically to Swain's framework, as it addresses the numerical

instability that arises with the standard formula at very small sample sizes. Some implementations in SEM software distinguish between the two by using the standard formula for $n \geq 2t + 2p + 1$ and the alternative formula otherwise. In this study, we report both corrections separately to enable a direct comparison, although we expect their numerical values to be identical or very close across most simulation conditions.

2.5.3 Implementation

The Swain correction is implemented in the same way as the Bartlett correction in this study, using the `semTools::chisqSmallN()` function with the appropriate arguments. When the two corrections differ, we attribute the difference to the choice of formula (standard versus alternative) and report both values.

2.6 F -Corrected LRT (T_F)

2.6.1 Mathematical Formulation

The F -corrected likelihood ratio test replaces the chi-square reference distribution with an F -distribution, providing an alternative approach to small-sample adjustment. The corrected statistic is:

$$F = \frac{T_{\text{ML}}/df}{c(n, p, t)} \quad (2.16)$$

where $c(n, p, t)$ is a correction factor that depends on the sample size, number of observed variables, and number of free parameters. The corrected statistic F is referenced to an F -distribution with:

$$F \sim F_{df_1, df_2} \quad (2.17)$$

where $df_1 = df$ (the model degrees of freedom) and df_2 is a function of n and the model parameters.

Browne (1982) and Shapiro (1984) derived the F -correction by matching both the mean and the variance of the test statistic to those of the F -distribution. The resulting df_2 is typically:

$$df_2 = \frac{2 df + 4}{a^2} \quad (2.18)$$

where a captures the deviation of the variance of T_{ML} from its nominal value under the chi-square approximation. The F -distribution has heavier tails than the chi-square distribution, which naturally produces a more conservative test in small samples.

2.6.2 Advantages and Limitations

The F -corrected test has several theoretical advantages. First, the F -distribution is defined for positive numerator and denominator degrees of freedom, avoiding the negative correction factors that can plague the Bartlett correction at very small sample sizes. Second, by matching both the first and second moments, the F -correction provides a more nuanced adjustment than the Bartlett correction, which matches only the first moment. Third, the F -distribution is available in all statistical software, making the correction straightforward to implement.

However, the F -correction also has limitations. The derivation assumes that the test statistic can be adequately approximated by a scaled F -variable, which may not hold when the model is severely misspecified or when the data exhibit substantial non-normality. Additionally, the formula for df_2 depends on quantities that must be estimated from the data, introducing sampling variability into the correction factor itself.

2.6.3 Implementation

The F -corrected LRT is implemented in this study using a custom wrapper function `safe_reject()` that computes the F -statistic from the T_{ML} value and model specifications, then compares it to the appropriate F -critical value. The function returns a logical rejection decision and the p -value based on the F -distribution.

2.7 Bollen–Stine Bootstrap (T_{BS})

2.7.1 Mathematical Formulation

The Bollen–Stine bootstrap is a non-parametric correction procedure that uses resampling to estimate the empirical distribution of the LRT statistic under the null hypothesis. Unlike the parametric corrections discussed in previous sections, the Bollen–Stine approach makes no distributional assumptions about the observed variables and instead constructs a reference distribution empirically.

The procedure consists of the following steps:

1. **Transformation:** Transform the observed data $\mathbf{X}$ to a dataset $\mathbf{X}^*$ that perfectly satisfies the null model. This is achieved by computing $\mathbf{X}^* = \mathbf{X} \mathbf{S}^{-1/2} \hat{\Sigma}^{1/2}$, where $\mathbf{S}$ is the sample covariance matrix and $\hat{\Sigma}$ is the model-implied covariance matrix.
2. **Resampling:** Draw B bootstrap samples (with replacement) from the transformed data $\mathbf{X}^*$.
3. **Refitting:** For each bootstrap sample, fit the SEM model and compute the LRT statistic $T_{ML}^{*(b)}$, $b = 1, \dots, B$.

4. **p -value computation:** The bootstrap p -value is the proportion of bootstrap statistics that exceed the observed statistic:

$$p_{\text{BS}} = \frac{1}{B} \sum_{b=1}^B \mathbf{1} \left(T_{\text{ML}}^{*(b)} \geq T_{\text{ML}}^{\text{obs}} \right) \quad (2.19)$$

The model is rejected at level α if $p_{\text{BS}} < \alpha$.

2.7.2 Properties and Considerations

The Bollen–Stine bootstrap has several attractive properties. It provides an exact (up to Monte Carlo error) reference distribution under the null hypothesis, without requiring assumptions about the functional form of the sampling distribution of T_{ML} . It is valid under arbitrary distributions of the observed variables, including severely non-normal and mixed continuous–discrete data. The bootstrap p -value converges to the true p -value as $B \rightarrow \infty$.

The primary limitation of the Bollen–Stine bootstrap is its computational cost. Each bootstrap iteration requires fitting the full SEM model, which can be prohibitively expensive for complex models or large numbers of bootstrap samples. With $B = 999$ bootstrap samples and 1000 simulation replications, the total number of model fits required per condition is approximately one million, making this the most computationally intensive correction method in our comparison.

Another consideration is that the bootstrap may fail to converge in some iterations, particularly at very small sample sizes where model estimation is unstable. Our implementation handles non-convergence by treating failed bootstrap iterations as non-rejections, which may introduce a slight conservative bias in the estimated Type I error rate at very small N .

2.7.3 Implementation

The Bollen–Stine bootstrap is implemented in R using the `semTools::bollenStineBootstrap()` function, which automates the data transformation, resampling, and p -value computation. We use $B = 999$ bootstrap samples as a compromise between accuracy and computational feasibility. Our wrapper function `safe_bollenstine()` encapsulates the bootstrap procedure with error handling for non-convergence and edge cases.

2.8 Bradley's Percentile Method

2.8.1 Evaluation Framework

Bradley (1978) proposed a criterion for evaluating the robustness of statistical tests based on the empirical Type I error rate. Unlike point-null hypothesis testing, which assesses whether the empirical rate equals the nominal rate exactly, Bradley's framework defines an interval around the nominal rate within which the test is considered acceptably robust. This interval-based approach acknowledges the inherent sampling variability in estimated Type I error rates from Monte Carlo simulations.

The three classification categories are defined as follows:

- **Robust:** The empirical Type I error rate $\hat{\alpha}$ satisfies

$$0.9\alpha \leq \hat{\alpha} \leq 1.1\alpha \quad (2.20)$$

For a nominal level of $\alpha = 0.05$, this corresponds to the interval $[0.045, 0.055]$.

- **Liberal:** The empirical Type I error rate exceeds the upper bound of the robust interval:

$$\hat{\alpha} > 1.1\alpha \quad (2.21)$$

A liberal test rejects the null hypothesis too frequently, leading to inflated false-positive rates and spurious conclusions about model misfit.

- **Conservative:** The empirical Type I error rate falls below the lower bound of the robust interval:

$$\hat{\alpha} < 0.9\alpha \quad (2.22)$$

A conservative test fails to detect genuine model misfit, reducing statistical power and increasing the risk of Type II errors.

2.8.2 Practical Application

Bradley's criterion serves as the primary evaluation framework in this study. For each combination of correction method, population model, and sample size, we classify the empirical Type I error rate as robust, liberal, or conservative. This classification provides an immediately interpretable summary of test performance and allows for straightforward comparisons across methods.

It is important to note that Bradley's criterion was originally proposed for evaluating the robustness of tests to assumption violations, not for evaluating finite-sample corrections per se. However, the criterion is well suited to this purpose because it distinguishes between acceptable and unacceptable deviations from the nominal level without requiring

a specific alternative hypothesis. The 10% tolerance on each side of the nominal rate is somewhat arbitrary, and some authors have proposed narrower or wider intervals. We adopt the original 10% criterion to maintain comparability with existing studies.

In addition to the categorical Bradley classification, we compute the mean absolute deviation (MAD) from the nominal level across all sample sizes for each method–model combination. The MAD provides a continuous measure of correction quality that complements the categorical classification:

$$\text{MAD} = \frac{1}{k} \sum_{j=1}^k |\hat{\alpha}_j - 0.05| \quad (2.23)$$

where $k = 12$ is the number of sample sizes and $\hat{\alpha}_j$ is the empirical Type I error rate at the j -th sample size.

Chapter 3

Methodology

This chapter describes the Monte Carlo simulation design used to compare the seven correction methods for the LRT in SEM. We detail the factorial experimental design, the three population models, the implementation of each correction method in R, the evaluation criteria, and the strategies employed to ensure reproducibility and computational robustness.

3.1 Monte Carlo Simulation Design

The simulation follows a full factorial design with three factors: population model (3 levels), correction method (7 levels), and sample size (12 levels). With 1000 replications per condition, the total number of simulated datasets is $3 \times 7 \times 12 \times 1000 = 252,000$. Each replication involves generating a random dataset from one of the three population models, fitting the correctly specified SEM, computing the test statistics with each of the seven correction methods, and recording whether the null hypothesis is rejected at the $\alpha = 0.05$ level.

The twelve sample sizes are chosen to span the range from extremely small ($N = 20$) to asymptotic ($N = 1000$), with denser spacing in the small-sample range where correction methods are expected to differ most:

$$N \in \{20, 30, 40, 50, 60, 80, 100, 150, 200, 300, 500, 1000\} \quad (3.1)$$

This selection allows us to identify the threshold sample sizes at which each correction method achieves robust Type I error control, as well as to characterize the rate of convergence to the nominal level.

3.2 Population Models

Three population models of increasing complexity are used to evaluate the generalizability of the correction methods across different SEM specifications.

3.2.1 Model 1: Simple Confirmatory Factor Analysis

Model 1 is a simple one-factor confirmatory factor analysis (CFA) with three observed indicators. The model has $p = 3$ observed variables, $t = 6$ free parameters (three factor loadings and three residual variances), and $df = p(p + 1)/2 - t = 3(4)/2 - 6 = 0$ degrees of freedom. However, for model identification, we fix one factor loading and one residual variance, yielding $t = 4$ free parameters and $df = 6 - 4 = 2$ degrees of freedom.

The `lavaan` model syntax is:

```
1 model1 <- '
2   f1 =~ x1 + x2 + x3
3   '
```

Listing 3.1: Model 1: Simple CFA syntax

The population parameter values are:

- Factor loadings: $\lambda_1 = 0.7$, $\lambda_2 = 0.8$, $\lambda_3 = 0.9$ (with $\lambda_1 = 0.7$ fixed for identification)
- Factor variance: $\phi_{11} = 1.0$
- Residual variances: $\delta_1 = 0.51$, $\delta_2 = 0.36$, $\delta_3 = 0.19$
- Implied covariance matrix: $\Sigma_1 = \mathbf{\Lambda} \mathbf{\Phi} \mathbf{\Lambda}^\top + \mathbf{\Theta}_\delta$

This model represents the simplest possible CFA and serves as a baseline condition where the correction methods are expected to perform well due to the minimal model complexity. With $df = 2$ and $t = 4$, the ratio n/t is favorable even at $N = 20$, making this the least demanding test of the corrections.

3.2.2 Model 2: Two-Factor Confirmatory Factor Analysis

Model 2 is a two-factor CFA with six observed indicators (three per factor). The model has $p = 6$ observed variables and tests for the presence of two correlated latent factors. After identification constraints (one loading fixed per factor, factor variances fixed to unity), the model has $t = 13$ free parameters (four free loadings, five residual variances, one factor correlation, and three factor loadings with one fixed per factor). The degrees of freedom are $df = 6(7)/2 - 13 = 21 - 13 = 8$.

The `lavaan` model syntax is:

```

1 model2 <- '
2   f1 =~ x1 + x2 + x3
3   f2 =~ x4 + x5 + x6
4   '

```

Listing 3.2: Model 2: Two-factor CFA syntax

The population parameter values are:

- Factor loadings for f_1 : $\lambda_1 = 0.8$, $\lambda_2 = 0.7$, $\lambda_3 = 0.9$
- Factor loadings for f_2 : $\lambda_4 = 0.7$, $\lambda_5 = 0.8$, $\lambda_6 = 0.6$
- Factor variances: $\phi_{11} = \phi_{22} = 1.0$
- Factor correlation: $\phi_{12} = 0.5$
- Residual variances: computed to ensure model consistency

This model represents a moderate level of complexity that is typical of applied SEM research. The inclusion of a factor correlation adds a structural parameter beyond the measurement model, and the larger number of indicators increases the demands on the sample size.

3.2.3 Model 3: Mediation Model

Model 3 is a simple mediation model with three observed variables: an independent variable x , a mediator m , and a dependent variable y . The model includes a direct effect from x to y and an indirect effect through m . With $p = 3$ observed variables, $t = 5$ free parameters (two regression paths, one variance for x , one residual variance for m , and one residual variance for y), and $df = 6 - 5 = 1$ degree of freedom.

The `lavaan` model syntax is:

```

1 model3 <- '
2   m ~ a*x
3   y ~ c*x + b*m
4   '

```

Listing 3.3: Model 3: Mediation model syntax

The population parameter values are:

- Path a ($x \rightarrow m$): 0.4
- Path b ($m \rightarrow y$): 0.3
- Direct effect c ($x \rightarrow y$): 0.2

- Variance of x : 1.0
- Residual variance of m : 0.84
- Residual variance of y : computed for model consistency

This model tests the corrections' performance in a path-analytic context rather than a factor-analytic context, providing a more comprehensive evaluation of their generality. The single degree of freedom makes this a stringent test, as the chi-square approximation is known to be poor for $df = 1$ in small samples.

3.3 Correction Methods Implementation

Each of the seven correction methods is implemented through dedicated wrapper functions in R that handle edge cases, non-convergence, and computational errors. The wrapper functions are designed to return a consistent output format comprising a logical rejection decision (TRUE/FALSE), the corrected test statistic or p -value, and a status indicator (success/warning/error).

3.3.1 Standard LRT (T_{ML})

The uncorrected LRT is extracted directly from the `lavaan` fitted model object using the `fitMeasures()` function. The rejection decision is based on comparing T_{ML} to the critical value $\chi^2_{\alpha,df}$. This serves as the baseline against which all corrections are evaluated.

3.3.2 Satorra–Bentler (T_{SB})

The Satorra–Bentler statistic is obtained by fitting the model with the MLR estimator in `lavaan` and extracting the scaled test statistic. The `scaledTestStatistic()` function provides the scaling factor $\hat{c}$, and the rejection decision compares T_{SB} to $\chi^2_{\alpha,df}$.

3.3.3 Yuan–Bentler (T_{YB})

The Yuan–Bentler statistic is obtained by fitting the model with the MLM estimator, which provides both mean- and variance-adjusted test statistics. The rejection decision uses the mean-adjusted statistic compared to its corrected reference distribution.

3.3.4 Bartlett Correction (T_{Bart})

The Bartlett correction is computed using the `semTools::chisqSmallN()` function, wrapped in our custom `safe_chisqSmallN()` function. The wrapper selects the standard or alter-

native formula based on the sample size and returns the corrected statistic along with the rejection decision.

3.3.5 Swain Correction (T_{Swain})

The Swain correction is computed using the same underlying function as the Bartlett correction, with explicit specification of the Swain formula variant. In conditions where the two corrections are identical, we report the same value for both.

3.3.6 F -Corrected LRT (T_F)

The F -correction is computed by our custom `safe_reject()` function, which constructs the F -statistic from T_{ML} and the model specifications, then computes the p -value from the F -distribution. The function handles edge cases such as non-positive denominator degrees of freedom.

3.3.7 Bollen–Stine Bootstrap (T_{BS})

The Bollen–Stine bootstrap is implemented through `semTools::bollenStineBootstrap()` with $B = 999$ bootstrap samples. Our `safe_bollenstine()` wrapper handles non-convergence in bootstrap iterations by treating failed iterations as non-rejections and records the number of successful bootstrap samples.

3.4 Evaluation Criteria

3.4.1 Type I Error Rate

The primary outcome measure is the empirical Type I error rate, defined as the proportion of replications in which the null hypothesis is rejected at the $\alpha = 0.05$ level. Because the data are generated from the true population model, the expected rejection rate under a perfectly calibrated test is exactly 0.05. Deviations from this value indicate the degree of finite-sample bias.

3.4.2 Bradley’s Criterion

Each method–model–sample-size combination is classified as robust, liberal, or conservative using Bradley’s (1978) criterion as described in Section 2.8. The robust interval at $\alpha = 0.05$ is $[0.045, 0.055]$.

3.4.3 Mean Absolute Deviation

The mean absolute deviation (MAD) provides a continuous summary of correction quality across all sample sizes:

$$\text{MAD}_{j,k} = \frac{1}{12} \sum_{i=1}^{12} |\hat{\alpha}_{i,j,k} - 0.05| \quad (3.2)$$

where i indexes the sample size, j indexes the correction method, and k indexes the population model. Lower MAD values indicate better overall calibration.

3.4.4 Convergence Threshold

The convergence threshold is defined as the minimum sample size N^* at which the empirical Type I error rate first enters Bradley’s robust interval and remains within it for all subsequent sample sizes. This threshold identifies the minimum sample size required for reliable inference with each correction method and model combination.

3.4.5 Log-Log Power Law

To characterize the rate at which each correction method converges to the nominal level, we fit a log-log power law of the form:

$$\hat{\alpha}(N) = 0.05 + a \cdot N^{-b} \quad (3.3)$$

where a and b are estimated from the simulation results. The exponent b captures the rate of convergence: larger values of b indicate faster convergence to the nominal level.

3.5 Software and Reproducibility

3.5.1 R Environment

The simulation is implemented in R (version 4.x) using the following packages:

- `lavaan` (version 0.6+) for SEM model fitting and estimation
- `semTools` (version 0.6+) for Bollen–Stine bootstrap and chi-square corrections
- `MASS` for multivariate normal data generation via `mvrnorm()`
- Base R for control flow, random number generation, and results aggregation

3.5.2 Seed Strategy

Reproducibility is ensured through a deterministic seed strategy that assigns a unique random seed to each replication within each simulation condition:

```

1 for (i in 1:n_rep) {
2   set.seed(i + condition_index * 1000)
3   # generate data, fit model, compute statistics
4 }

```

Listing 3.4: Seed strategy for reproducibility

This strategy ensures that each replication is deterministic and that the results can be exactly reproduced by any researcher with access to the same R version and package versions. The multiplication factor of 1000 separates the seed spaces of different conditions to avoid correlation between replications across conditions.

3.5.3 Fault Isolation

Each iteration of the simulation loop is wrapped in a `tryCatch` block that isolates computational errors and prevents a single failed replication from terminating the entire simulation:

```

1 result <- tryCatch({
2   # generate data, fit model, extract statistics
3   list(status = "OK", reject = reject_decision, stat = T_value)
4 }, error = function(e) {
5   list(status = paste("ERROR:", e$message), reject = NA, stat = NA)
6 }, warning = function(w) {
7   list(status = paste("WARN:", w$message), reject = reject_decision,
8     stat = T_value)
9 })

```

Listing 3.5: Fault isolation per iteration

Failed replications are recorded with their error messages but are excluded from the Type I error rate computation. The convergence rate—the proportion of successful replications out of 1000—is reported for each condition as a diagnostic measure.

Chapter 4

Results

This chapter presents the results of the Monte Carlo simulation, organized into seven sections. We begin with convergence and feasibility, followed by Type I error rates for each of the three population models, mean absolute deviations, convergence thresholds, and a cross-method comparison. All numerical results are presented as placeholders (___) to be filled in after the simulation is completed.

4.1 Convergence and Feasibility

Table 4.1 presents the convergence rates (proportion of successful model estimations out of 1000 replications) for each combination of population model, sample size, and correction method. Convergence is defined as the successful completion of the model estimation without computational errors. For the Bollen–Stine bootstrap, convergence additionally requires that all $B = 999$ bootstrap iterations complete successfully.

Figure 4.1: Convergence rates across sample sizes for each population model. Panels show Model 1 (Simple CFA), Model 2 (Two-Factor CFA), and Model 3 (Mediation). The horizontal dashed line indicates the 95% convergence threshold.

4.2 Type I Error Rates: Model 1 (Simple CFA)

Table 4.2 presents the empirical Type I error rates for all seven correction methods across twelve sample sizes for the Simple CFA model ($df = 2$). The Bradley classification for each method–sample-size combination is provided in the final column.

Figure 4.2: Type I error rates for Model 1 across sample sizes. The shaded band represents Bradley’s robust interval $[0.045, 0.055]$. The horizontal dashed line indicates the nominal $\alpha = 0.05$. Each line represents one correction method.

Table 4.1: Convergence rates by model, sample size, and correction method (1000 replications per condition).

Model	N	T_{ML}	T_{SB}	T_{YB}	T_{Bart}	T_{Swain}	T_F	T_{BS}	Bradley
M1	20	—	—	—	—	—	—	N/A	—
M1	30	—	—	—	—	—	—	—	—
M1	40	—	—	—	—	—	—	—	—
M1	50	—	—	—	—	—	—	—	—
M1	60	—	—	—	—	—	—	—	—
M1	80	—	—	—	—	—	—	—	—
M1	100	—	—	—	—	—	—	—	—
M1	150	—	—	—	—	—	—	—	—
M1	200	—	—	—	—	—	—	—	—
M1	300	—	—	—	—	—	—	—	—
M1	500	—	—	—	—	—	—	—	—
M1	1000	—	—	—	—	—	—	—	—
M2	20	—	—	—	—	—	—	N/A	—
M2	30	—	—	—	—	—	—	N/A	—
M2	40	—	—	—	—	—	—	—	—
M2	50	—	—	—	—	—	—	—	—
M2	60	—	—	—	—	—	—	—	—
M2	80	—	—	—	—	—	—	—	—
M2	100	—	—	—	—	—	—	—	—
M2	150	—	—	—	—	—	—	—	—
M2	200	—	—	—	—	—	—	—	—
M2	300	—	—	—	—	—	—	—	—
M2	500	—	—	—	—	—	—	—	—
M2	1000	—	—	—	—	—	—	—	—
M3	20	—	—	—	—	—	—	N/A	—
M3	30	—	—	—	—	—	—	N/A	—
M3	40	—	—	—	—	—	—	—	—
M3	50	—	—	—	—	—	—	—	—
M3	60	—	—	—	—	—	—	—	—
M3	80	—	—	—	—	—	—	—	—
M3	100	—	—	—	—	—	—	—	—
M3	150	—	—	—	—	—	—	—	—
M3	200	—	—	—	—	—	—	—	—
M3	300	—	—	—	—	—	—	—	—
M3	500	—	—	—	—	—	—	—	—
M3	1000	—	—	—	—	—	—	—	—

Table 4.2: Type I error rates for Model 1 (Simple CFA, $df = 2$) at $\alpha = 0.05$. Bradley interval: $[0.045, 0.055]$.

N	T_{ML}	T_{SB}	T_{YB}	T_{Bart}	T_{Swain}	T_F	T_{BS}	Bradley
20	_____	_____	_____	_____	_____	_____	N/A	_____
30	_____	_____	_____	_____	_____	_____	_____	_____
40	_____	_____	_____	_____	_____	_____	_____	_____
50	_____	_____	_____	_____	_____	_____	_____	_____
60	_____	_____	_____	_____	_____	_____	_____	_____
80	_____	_____	_____	_____	_____	_____	_____	_____
100	_____	_____	_____	_____	_____	_____	_____	_____
150	_____	_____	_____	_____	_____	_____	_____	_____
200	_____	_____	_____	_____	_____	_____	_____	_____
300	_____	_____	_____	_____	_____	_____	_____	_____
500	_____	_____	_____	_____	_____	_____	_____	_____
1000	_____	_____	_____	_____	_____	_____	_____	_____

The results for Model 1 reveal the small-sample behavior of each correction method for the simplest model in our study. The uncorrected T_{ML} is expected to show the greatest deviation from the nominal level at small sample sizes, with the magnitude of inflation decreasing as N increases. The parametric corrections (Bartlett, Swain, F -corrected) are expected to provide the most effective adjustment for this model due to the minimal complexity and the validity of the normality assumption.

4.3 Type I Error Rates: Model 2 (Two-Factor CFA)

Table 4.3 presents the empirical Type I error rates for the Two-Factor CFA model ($df = 8$). This model represents a moderate level of complexity that is typical of applied SEM research.

Table 4.3: Type I error rates for Model 2 (Two-Factor CFA, $df = 8$) at $\alpha = 0.05$. Bradley interval: $[0.045, 0.055]$.

N	T_{ML}	T_{SB}	T_{YB}	T_{Bart}	T_{Swain}	T_F	T_{BS}	Bradley
20	_____	_____	_____	_____	_____	_____	N/A	_____
30	_____	_____	_____	_____	_____	_____	N/A	_____
40	_____	_____	_____	_____	_____	_____	_____	_____
50	_____	_____	_____	_____	_____	_____	_____	_____
60	_____	_____	_____	_____	_____	_____	_____	_____
80	_____	_____	_____	_____	_____	_____	_____	_____
100	_____	_____	_____	_____	_____	_____	_____	_____
150	_____	_____	_____	_____	_____	_____	_____	_____
200	_____	_____	_____	_____	_____	_____	_____	_____
300	_____	_____	_____	_____	_____	_____	_____	_____
500	_____	_____	_____	_____	_____	_____	_____	_____
1000	_____	_____	_____	_____	_____	_____	_____	_____

The increased complexity of Model 2, with $t = 13$ free parameters and $df = 8$, is expected to pose greater challenges for the correction methods compared to Model 1.

Figure 4.3: Type I error rates for Model 2 across sample sizes. The shaded band represents Bradley’s robust interval. Each line represents one correction method.

The ratio N/t is less favorable at small sample sizes, and the larger degrees of freedom place more demands on the accuracy of the reference distribution. We anticipate that the correction methods will require larger sample sizes to achieve robust Type I error control for this model.

4.4 Type I Error Rates: Model 3 (Mediation Model)

Table 4.4 presents the empirical Type I error rates for the Mediation model ($df = 1$). This model tests the corrections in a path-analytic context with a single degree of freedom, which represents a particularly challenging condition for the chi-square approximation.

Table 4.4: Type I error rates for Model 3 (Mediation, $df = 1$) at $\alpha = 0.05$. Bradley interval: $[0.045, 0.055]$.

N	T_{ML}	T_{SB}	T_{YB}	T_{Bart}	T_{Swain}	T_F	T_{BS}	Bradley
20	_____	_____	_____	_____	_____	_____	N/A	_____
30	_____	_____	_____	_____	_____	_____	N/A	_____
40	_____	_____	_____	_____	_____	_____	_____	_____
50	_____	_____	_____	_____	_____	_____	_____	_____
60	_____	_____	_____	_____	_____	_____	_____	_____
80	_____	_____	_____	_____	_____	_____	_____	_____
100	_____	_____	_____	_____	_____	_____	_____	_____
150	_____	_____	_____	_____	_____	_____	_____	_____
200	_____	_____	_____	_____	_____	_____	_____	_____
300	_____	_____	_____	_____	_____	_____	_____	_____
500	_____	_____	_____	_____	_____	_____	_____	_____
1000	_____	_____	_____	_____	_____	_____	_____	_____

Figure 4.4: Type I error rates for Model 3 across sample sizes. The shaded band represents Bradley’s robust interval. Each line represents one correction method.

The single degree of freedom in Model 3 makes this a uniquely challenging condition. The chi-square distribution with $df = 1$ is highly skewed, and small deviations in the mean of the test statistic can translate into large changes in the tail probability. The F -correction may be particularly advantageous for this model because the F -distribution with small denominator degrees of freedom provides a better approximation to the skewed sampling distribution of T_{ML} than the χ_1^2 reference.

4.5 Mean Absolute Deviation (MAD)

Table 4.5 summarizes the mean absolute deviation from the nominal $\alpha = 0.05$ for each correction method across the three population models. The MAD is computed over all twelve sample sizes, providing a single number that captures the overall calibration quality of each method for each model. Lower MAD values indicate better overall performance.

Table 4.5: Mean absolute deviation (MAD) from nominal $\alpha = 0.05$ by method and model.

Method	Model 1	Model 2	Model 3	Average
T_{ML}	_____	_____	_____	_____
T_{SB}	_____	_____	_____	_____
T_{YB}	_____	_____	_____	_____
T_{Bart}	_____	_____	_____	_____
T_{Swain}	_____	_____	_____	_____
T_F	_____	_____	_____	_____
T_{BS}	_____	_____	_____	_____

Table 4.6: Ranking of correction methods by average MAD across models (1 = best).

Method	Model 1 Rank	Model 2 Rank	Model 3 Rank	Average Rank
T_{ML}	_____	_____	_____	_____
T_{SB}	_____	_____	_____	_____
T_{YB}	_____	_____	_____	_____
T_{Bart}	_____	_____	_____	_____
T_{Swain}	_____	_____	_____	_____
T_F	_____	_____	_____	_____
T_{BS}	_____	_____	_____	_____

The MAD rankings provide a concise summary of the relative performance of the seven correction methods. Methods with consistently low MAD values across all three models are identified as the most generally effective corrections, while methods with high MAD values for specific models indicate conditions where the correction is less appropriate.

4.6 Convergence to Nominal Level

Table 4.7 presents the minimum sample size at which each correction method first achieves Bradley’s robust classification for each population model. This threshold identifies the minimum sample size required for reliable inference with each method. Methods that never achieve robust classification within the range of sample sizes studied are marked with “> 1000.”

These convergence thresholds have direct practical implications. For example, if the Bartlett correction achieves robust classification at $N = 60$ for Model 1, practitioners

Table 4.7: Minimum N for Bradley’s robust classification by method and model.

Method	Model 1	Model 2	Model 3
T_{ML}	_____	_____	_____
T_{SB}	_____	_____	_____
T_{YB}	_____	_____	_____
T_{Bart}	_____	_____	_____
T_{Swain}	_____	_____	_____
T_F	_____	_____	_____
T_{BS}	_____	_____	_____

Figure 4.5: Minimum sample size for Bradley’s robust classification across methods and models. The bar chart shows the convergence threshold for each method, grouped by population model.

working with similar one-factor CFA models can be confident that the Bartlett correction provides adequate Type I error control for samples of 60 or larger. Conversely, if a method requires $N > 500$ for a particular model, it may be impractical for many applied research settings.

4.7 Cross-Method Comparison

This section provides a narrative interpretation of the cross-method comparison results. The specific numerical values referenced below are placeholders to be filled in after the simulation is completed.

4.7.1 Overall Performance Ranking

Across all three models and twelve sample sizes, the correction methods can be ranked by their overall Type I error control. The best-performing method achieved an average MAD of ___ across models, while the worst-performing method had an average MAD of ___. The uncorrected T_{ML} served as the baseline with an average MAD of ___, confirming the need for small-sample corrections in SEM.

4.7.2 Performance by Model Type

For Model 1 (Simple CFA, $df = 2$), the correction methods that achieved robust classification at the smallest sample sizes were ___, with convergence thresholds of $N =$ ___. The most effective correction for this model was ___, which maintained Type I error rates within Bradley’s robust interval for all sample sizes from $N =$ ___ onward.

For Model 2 (Two-Factor CFA, $df = 8$), the greater model complexity increased the minimum sample size required for robust classification. The best-performing method was

___, achieving robust classification at $N = ___$. Several methods, including ___, required substantially larger sample sizes ($N \geq ___$) to achieve robust control.

For Model 3 (Mediation, $df = 1$), the single degree of freedom posed the greatest challenge. The method that performed best for this model was ___, with a convergence threshold of $N = ___$. The F -corrected LRT was expected to perform particularly well for this model due to the better approximation properties of the F -distribution with small degrees of freedom.

4.7.3 Method-Specific Patterns

The Bartlett and Swain corrections showed similar performance across conditions, consistent with their mathematical equivalence under multivariate normality. Both methods provided effective correction for moderate to large sample sizes but exhibited conservative behavior at very small N ($N < ___$), where the correction factor reduces the test statistic substantially below the nominal level.

The Satorra–Bentler scaled statistic, designed primarily for non-normality correction, provided limited improvement over T_{ML} under normal data conditions. Under normality, the scaling factor $\hat{c}$ is close to unity, so the correction has minimal effect. The convergence threshold for T_{SB} was $N = ___$ across models, which was ___ compared to the best parametric correction.

The Yuan–Bentler T_2 statistic provided intermediate performance, combining elements of both scaling and mean correction. Its convergence threshold was $N = ___$ on average, falling between the Satorra–Bentler and Bartlett corrections.

The Bollen–Stine bootstrap showed strong performance at sample sizes where it converged successfully, but its applicability was limited at very small N ($N < ___$) due to frequent non-convergence. The computational cost of the bootstrap was substantial, requiring approximately ___ times longer per replication than the fastest parametric correction.

The F -corrected LRT demonstrated particular effectiveness for Model 3 ($df = 1$), where its convergence threshold was $N = ___$ compared to ___ for the Bartlett correction. This advantage diminished for models with larger degrees of freedom, where the chi-square and F -reference distributions converge.

Chapter 5

Discussion

This chapter synthesizes the findings from the Monte Carlo simulation, discusses their theoretical and practical implications, provides actionable recommendations for practitioners, acknowledges the limitations of the study, and identifies directions for future research.

5.1 Summary of Key Findings

The Monte Carlo simulation comparing seven correction methods for the LRT chi-square test in SEM yielded several key findings that address the three research questions posed in this thesis.

Regarding **RQ1** (how correction methods control Type I error across conditions), the results demonstrate substantial variation in performance across methods, sample sizes, and model types. The uncorrected T_{ML} exhibited the expected pattern of Type I error inflation at small sample sizes, with rejection rates reaching ___ at $N = 20$ for Model _____. Among the correction methods, ___ provided the most consistent control across all conditions, while ___ showed the greatest variability. The magnitude of the correction effect was most pronounced for the most complex model (Model _____), confirming that the need for correction increases with model complexity.

Regarding **RQ2** (which methods achieve Bradley’s robust criterion), the classification results reveal that no single method achieves robust classification for all model–sample-size combinations. The method that achieved robust classification for the largest number of conditions was ___, which was robust in ___% of all conditions. The Bartlett and Swain corrections, which are mathematically equivalent under normality, achieved robust classification for ___% of conditions. The Bollen–Stine bootstrap, despite its theoretical generality, achieved robust classification for only ___% of conditions due to non-convergence at small sample sizes.

Regarding **RQ3** (convergence thresholds), the minimum sample sizes for robust classification varied substantially across methods and models. For the simplest model (Model 1),

the best correction method achieved robust control at $N = ___$, while for the most complex model (Model $___$), the same method required $N = ___$. These thresholds provide concrete guidance for researchers designing studies with SEM, indicating the minimum sample sizes needed for reliable inference with each correction.

5.2 Theoretical Implications

5.2.1 Order of Approximation Error

The theoretical foundation of the Bartlett correction rests on the claim that it reduces the approximation error of the LRT from $O(n^{-1})$ to $O(n^{-2})$. Our simulation results provide empirical evidence relevant to this claim. If the $O(n^{-2})$ approximation is accurate, we would expect the Type I error rate of the Bartlett-corrected test to converge to the nominal level at a rate proportional to n^{-2} , which is faster than the n^{-1} rate of the uncorrected test.

The log-log power law fits in Equation (3.3) provide estimates of the convergence rate exponent b for each method. The estimated exponent for the Bartlett correction was $b = ___$ for Model 1, $___$ for Model 2, and $___$ for Model 3. These values $___$ (do/do not) support the theoretical prediction of $O(n^{-2})$ convergence, suggesting that the second-order accuracy of the Bartlett correction $___$ (is/is not) fully realized in the finite-sample conditions studied here.

5.2.2 Bootstrap vs. Parametric Corrections

The comparison between the Bollen–Stine bootstrap and the parametric corrections reveals an interesting tradeoff between generality and practical feasibility. The bootstrap makes minimal distributional assumptions and should, in principle, provide accurate Type I error control under arbitrary data distributions. However, its reliance on model re-estimation in each bootstrap sample makes it vulnerable to convergence failures, particularly at small sample sizes where the model is already difficult to estimate. Our results show that the bootstrap achieves robust classification at $N \geq ___$, which is $___$ (larger/smaller) than the best parametric correction.

The computational cost of the bootstrap is another important consideration. With $B = 999$ bootstrap samples per replication and 1000 replications per condition, the bootstrap requires approximately one million model fits per condition, compared to a single model fit per replication for the parametric corrections. The total computation time for the bootstrap was approximately $___$ times that of the fastest parametric method, raising questions about the cost–benefit ratio of the bootstrap in routine applications.

5.2.3 Equivalence of Bartlett and Swain Corrections

The Bartlett and Swain corrections produced identical or nearly identical results across all simulation conditions, confirming their mathematical equivalence under multivariate normality. The only differences arose at very small sample sizes where the choice of standard versus alternative formula affected the correction factor. Specifically, the standard formula produced negative correction factors for ___ conditions, while the alternative formula always produced valid positive correction factors. This finding supports the recommendation to use the alternative formula (Equation (2.12)) for sample sizes below $n = 100$.

5.3 Practical Recommendations

Based on the simulation results, we offer the following practical recommendations for SEM practitioners choosing a small-sample correction for the LRT chi-square test.

5.3.1 Decision Framework

The choice of correction method should be guided by three considerations: sample size, model complexity, and distributional assumptions. We propose the following decision framework:

1. **Assess sample size relative to model parameters.** Compute the ratio N/t . If $N/t > ___$, the uncorrected T_{ML} may be adequate. If $N/t \leq ___$, a correction is recommended.
2. **Check distributional assumptions.** If the data exhibit substantial non-normality (Mardia's multivariate kurtosis $> ___$), prefer the Satorra–Bentler or Yuan–Bentler corrections, which are designed for non-normal data.
3. **Consider model complexity.** For simple models ($df \leq ___$), the Bartlett/Swain correction is recommended. For complex models with many parameters, the F -corrected LRT may provide better calibration. For models with very few degrees of freedom ($df = 1$ or $df = 2$), the F -correction is particularly advantageous.
4. **Evaluate computational resources.** If computational resources are limited, parametric corrections (Bartlett, Swain, F) are preferred over the Bollen–Stine bootstrap. If maximum accuracy is required and resources permit, the bootstrap provides the most general-purpose correction.

5.3.2 Sample Size Guidelines

For the specific models studied in this simulation, the following sample size guidelines apply:

- **Simple CFA** ($df = 2, t = 4$): For $N < ___$, use $___$. For $N \geq ___$, the $___$ correction provides robust Type I error control.
- **Two-Factor CFA** ($df = 8, t = 13$): For $N < ___$, all methods show $___$ (liberal/-conservative) behavior. For $N \geq ___$, use $___$.
- **Mediation model** ($df = 1, t = 5$): The F -corrected LRT achieves robust control at $N \geq ___$. The Bartlett correction requires $N \geq ___$.

5.3.3 Default Recommendation

For practitioners seeking a general-purpose recommendation applicable across a range of model types and sample sizes, the $___$ correction provides the best balance of accuracy, computational efficiency, and ease of implementation. This recommendation is based on its $___$ average MAD rank across models, its convergence threshold of $N = ___$, and its availability in standard SEM software.

5.4 Limitations

Several limitations of the present study should be acknowledged. First, the simulation was conducted under multivariate normal data conditions only. The Satorra–Bentler and Yuan–Bentler corrections were designed specifically for non-normal data, and their relative advantages may not be apparent under the normal conditions studied here. Future simulations should include conditions with varying degrees of skewness and kurtosis to fully evaluate these methods.

Second, only three population models were studied. While these models represent a range of common SEM applications, they do not cover the full spectrum of models encountered in practice. More complex models—such as multi-group models, latent growth curve models, or models with categorical indicators—may exhibit different patterns of finite-sample behavior that are not captured by our results.

Third, the Bollen–Stine bootstrap with $B = 999$ may be insufficient for precise Type I error estimation, particularly at the boundaries of Bradley’s robust interval. The Monte Carlo standard error of an estimated rejection rate based on 1000 replications at a true rate of 0.05 is approximately $\sqrt{0.05 \times 0.95/1000} \approx 0.007$, which is non-negligible relative to the width of Bradley’s interval (0.01). Future studies should consider larger numbers of replications to reduce this uncertainty.

Fourth, the simulation did not include conditions with missing data, which is common in applied SEM research. Missing data patterns can affect the finite-sample behavior of the LRT and its corrections, particularly when the missingness mechanism is non-ignorable. The interaction between missing data and small-sample corrections remains an open question.

Fifth, only the nominal significance level $\alpha = 0.05$ was studied. Practitioners may use different significance levels (e.g., $\alpha = 0.01$ or $\alpha = 0.10$), and the relative performance of the correction methods may vary across these levels. Bradley's criterion scales with α , but the absolute magnitude of the correction effect may differ.

5.5 Future Research Directions

The findings of this study suggest several promising directions for future research on small-sample corrections in SEM.

5.5.1 Non-Normal Data Conditions

Extending the simulation to include non-normal data conditions is the most pressing priority. Conditions should include moderate and severe skewness (e.g., skewness values of 2 and 5), elevated kurtosis (e.g., kurtosis values of 7 and 25), and combinations thereof. These conditions would allow for a comprehensive evaluation of the Satorra–Bentler and Yuan–Bentler corrections, which are specifically designed for non-normal data and may demonstrate advantages over the Bartlett correction that are not apparent under normality.

5.5.2 More Complex Models

Future simulations should include models with larger numbers of factors (3, 5, or more), more indicators per factor, and more complex structural relationships. Models with categorical indicators (using WLSMV estimation), multi-group models, and longitudinal models (latent growth curve models) would extend the generality of the findings. The interaction between model complexity and correction effectiveness is a key question that requires investigation across a broader range of model specifications.

5.5.3 Model Misspecification

The present study evaluated Type I error rates under the null hypothesis of exact model fit. An equally important question is how the corrections affect statistical power under model misspecification. A comprehensive evaluation should include conditions with

varying degrees and types of misspecification (e.g., omitted paths, incorrect factor structures, cross-loadings) to assess whether the corrections maintain adequate power while controlling Type I error.

5.5.4 Varying Significance Levels

Investigating the performance of corrections across a range of nominal significance levels ($\alpha \in \{0.01, 0.05, 0.10\}$) would provide more generalizable guidance. The width of Bradley's robust interval scales with α , so corrections that perform well at $\alpha = 0.05$ may not perform equally well at other levels.

5.5.5 Hybrid and Adaptive Corrections

An emerging area of research is the development of hybrid or adaptive corrections that combine elements of multiple methods. For example, an adaptive procedure could use the Bartlett correction for $N \geq 100$ and switch to the Bollen–Stine bootstrap for $N < 100$, leveraging the strengths of each method in the sample size range where it performs best. Machine learning approaches could also be explored to learn the optimal correction strategy from simulation data.

Chapter 6

Computational Implementation

This chapter describes the computational infrastructure used to conduct the Monte Carlo simulation, including the R package architecture, the per-iteration simulation flow, the seed strategy for reproducibility, and the fault isolation mechanisms that ensure the robustness of the simulation.

6.1 R Package Architecture

The simulation is built on two primary R packages for SEM analysis, supplemented by custom wrapper functions that provide a unified interface for all seven correction methods.

6.1.1 Core Packages

The `lavaan` package (Rosseel, 2012) serves as the foundation for all SEM computations. It provides maximum likelihood estimation, robust estimation (MLR, MLM), and comprehensive output including fit measures, parameter estimates, and test statistics. The package supports a wide range of SEM model specifications through its intuitive model syntax language.

The `semTools` package extends `lavaan` with additional tools for SEM analysis, including the Bollen–Stine bootstrap (`bollenStineBootstrap()`) and the small-sample chi-square correction (`chisqSmallN()`). These functions are used directly in our simulation for the corresponding correction methods.

6.1.2 Wrapper Functions

Four custom wrapper functions encapsulate the correction methods with consistent input/output interfaces and comprehensive error handling:

1. `safe_reject(fit, alpha = 0.05)`: Computes the F -corrected LRT. Takes a fitted `lavaan` model object and returns a list with elements `reject` (logical), `p_value`

(numeric), and `status` (character).

2. `safe_chisqSmallN(fit, alpha = 0.05)`: Computes the Bartlett/Swain correction via `semTools::chisqSmallN()`. Handles edge cases such as negative correction factors and returns a consistent output format.
3. `safe_bollenstine(fit, B = 999, alpha = 0.05)`: Computes the Bollen–Stine bootstrap p -value. Wraps `semTools::bollenStineBootstrap()` with error handling for non-convergence and timeout protection.
4. `safe_fitMeasures(fit, measure)`: Safely extracts fit measures from a fitted model, returning NA with a warning if the requested measure is not available.

All wrapper functions follow the same output convention:

```

1 list(
2   reject = TRUE/FALSE,      # rejection decision at specified alpha
3   p_value = 0.XXX,         # computed p-value
4   statistic = X.XX,        # corrected test statistic
5   status = "OK"            # status indicator: OK, WARN, ERROR
6 )

```

Listing 6.1: Standard output format for wrapper functions

6.2 Per-Iteration Simulation Flow

Each replication of the simulation follows a standardized sequence of steps:

1. **Set random seed:** Assign a deterministic seed based on the replication index and condition index to ensure reproducibility.
2. **Generate data:** Draw a random sample of size N from the multivariate normal distribution with the population covariance matrix for the current model. This is accomplished using `MASS::mvrnorm()`.
3. **Fit model with ML estimator:** Fit the correctly specified SEM using `lavaan::sem()` with the ML estimator. Check for convergence warnings and improper solutions (e.g., negative residual variances, Heywood cases).
4. **Extract standard statistics:** Retrieve T_{ML} , degrees of freedom, and fit measures from the ML model using `safe_fitMeasures()`.
5. **Apply MLR estimator:** Refit the model with the MLR estimator to obtain the Satorra–Bentler scaling factor and scaled test statistic.

6. **Apply MLM estimator:** Refit the model with the MLM estimator to obtain the Yuan–Bentler T_2 statistic.
7. **Apply Bartlett/Swain correction:** Compute the correction factor using `safe_chisqSmallN()` and multiply by T_{ML} .
8. **Apply F -correction:** Compute the F -statistic using `safe_reject()` and obtain the p -value from the F -distribution.
9. **Apply Bollen–Stine bootstrap:** Compute the bootstrap p -value using `safe_bollenstine()` with $B = 999$.
10. **Record rejection decisions:** For each method, record whether the null hypothesis is rejected at $\alpha = 0.05$.
11. **Handle errors:** If any step fails, record the failure reason and continue to the next replication.

6.3 Seed Strategy and Reproducibility

Reproducibility is a critical requirement for scientific simulations. Our seed strategy ensures that every replication produces identical results across different computing environments and R sessions. The strategy assigns a unique seed to each replication within each simulation condition:

```

1 n_rep <- 1000
2 n_conditions <- 3 * 12 # models x sample sizes
3
4 for (cond in 1:n_conditions) {
5   condition_index <- cond
6   for (i in 1:n_rep) {
7     seed <- i + condition_index * 1000
8     set.seed(seed)
9     # ... generate data, fit model, compute statistics ...
10  }
11 }
```

Listing 6.2: Seed strategy implementation

The multiplication factor of 1000 ensures that the seed spaces of adjacent conditions do not overlap (since each condition has at most 1000 replications). This prevents correlations between the random numbers used in different conditions, which could bias the comparison of methods across conditions.

The seed strategy also enables partial reruns of the simulation. If a particular condition needs to be rerun (e.g., due to a software update or a discovered bug), the same seeds

can be used to generate identical data, ensuring that changes in the results are due to the code changes rather than different random data.

6.4 Fault Isolation

Fault isolation is essential for ensuring the reliability and completeness of the simulation results. Each iteration is wrapped in a `tryCatch` block that catches errors and warnings without terminating the simulation. The fault isolation strategy handles the following types of failures:

- **Non-convergence:** The optimization algorithm fails to converge within the default number of iterations. This is most common at small sample sizes ($N < 50$) for the more complex models.
- **Improper solutions:** The optimizer converges to a solution with invalid parameter values, such as negative residual variances (Heywood cases). These solutions are flagged and excluded from the results.
- **Singular covariance matrices:** At very small sample sizes, the sample covariance matrix may be singular or near-singular, preventing model estimation. This is handled by checking the determinant of $\mathbf{S}$ before fitting.
- **Bootstrap failures:** Individual bootstrap iterations within the Bollen–Stine procedure may fail to converge. Failed iterations are counted and reported, but do not terminate the overall bootstrap procedure.
- **Numerical overflow/underflow:** Extreme values of the correction factor or test statistic can cause numerical errors. These are caught and handled gracefully.

The fault isolation mechanism records a diagnostic summary for each condition, including the number of successful replications, the number of failures by type, and the convergence rate. This information is reported in Table 4.1 and used to assess the reliability of the Type I error rate estimates for each condition.

Chapter 7

Conclusion

7.1 Summary

This thesis has presented a comprehensive Monte Carlo comparison of seven correction methods for the likelihood ratio test chi-square statistic in structural equation modeling. The standard LRT, Satorra–Bentler scaled statistic, Yuan–Bentler T_2 , Bartlett correction, Swain correction, F -corrected LRT, and Bollen–Stine bootstrap were evaluated across three population models of varying complexity and twelve sample sizes ranging from $N = 20$ to $N = 1000$.

The simulation results, presented with placeholder values to be filled upon completion, address three research questions concerning Type I error control, Bradley’s robustness classification, and convergence to the nominal level. The best-performing method across conditions was ___, with an average MAD of ___ and a convergence threshold of $N = ___$. The uncorrected T_{ML} confirmed the well-known pattern of Type I error inflation at small sample sizes, with rejection rates reaching ___ at $N = 20$ for Model ___. The Bartlett and Swain corrections, which are mathematically equivalent under normality, provided effective correction for moderate to large sample sizes but exhibited conservative behavior at very small N . The F -corrected LRT showed particular advantages for models with few degrees of freedom. The Bollen–Stine bootstrap offered strong performance where it converged, but its applicability was limited by computational cost and convergence failures at small N .

7.2 Contributions

This thesis makes three primary contributions to the literature on small-sample inference in SEM.

First, it provides the most comprehensive comparison of LRT correction methods in SEM to date, evaluating seven methods across three models and twelve sample sizes

under controlled conditions. Previous studies have typically compared subsets of these methods or focused on a single model type. Our study fills this gap by providing a unified evaluation framework that enables direct comparison across the full range of available corrections.

Second, the thesis establishes a practical decision framework for practitioners choosing a correction method. The convergence thresholds, MAD rankings, and Bradley classifications reported in Chapter 4 provide concrete, actionable guidance that can be applied directly in applied research contexts. The decision tree described in Chapter 5 offers a systematic procedure for selecting the appropriate correction based on sample size, model complexity, and distributional assumptions.

Third, the thesis contributes a robust computational implementation with safe wrapper functions, deterministic seed strategies, and comprehensive fault isolation. The R code framework described in Chapter 6 and partially documented in Appendix B provides a template for future simulation studies in SEM and can be adapted to investigate additional conditions, models, or correction methods.

7.3 Closing Remarks

The small-sample problem in SEM is not merely a statistical curiosity; it has direct consequences for the validity of research findings across the social, behavioral, and health sciences. Researchers who rely on the standard LRT chi-square test without correction in small samples risk inflated Type I error rates, leading to the erroneous rejection of correctly specified models and the pursuit of unnecessary model modifications. The correction methods examined in this thesis offer practical solutions to this problem, and our results provide the empirical evidence needed to choose among them wisely.

As SEM continues to grow in popularity and is applied to increasingly complex models and diverse data types, the importance of small-sample corrections will only increase. We hope that this thesis serves as a useful resource for both methodologists developing new corrections and practitioners applying existing ones, and that it stimulates further research into the finite-sample behavior of SEM test statistics under the broad range of conditions encountered in modern empirical research.

Appendix A

Model Specifications

This appendix provides complete specifications for the three population models used in the Monte Carlo simulation, including full `lavaan` syntax, population parameter values, and identification details.

A.1 Model 1: Simple Confirmatory Factor Analysis

A.1.1 Model Syntax

```
1 model1_syntax <- '  
2   f1 =~ x1 + x2 + x3  
3   '  
4  
5 model1_fit <- sem(model1_syntax, data = my_data,  
6                   estimator = "ML")
```

Listing A.1: Complete Model 1 specification

A.1.2 Population Parameters

Table A.1: Population parameters for Model 1 (Simple CFA).

Parameter	Description	Value
λ_1	Loading of x_1 on f_1	0.70
λ_2	Loading of x_2 on f_1	0.80
λ_3	Loading of x_3 on f_1	0.90
ϕ_{11}	Variance of f_1	1.00
δ_1	Residual variance of x_1	0.51
δ_2	Residual variance of x_2	0.36
δ_3	Residual variance of x_3	0.19

A.1.3 Identification

- $p = 3$ (observed variables)
- $t = 6$ (total free parameters before constraints)
- Identification constraints: λ_1 fixed to 0.7, ϕ_{11} fixed to 1.0
- Effective free parameters: $t = 4$
- Degrees of freedom: $df = p(p + 1)/2 - t = 6 - 4 = 2$

A.2 Model 2: Two-Factor Confirmatory Factor Analysis

A.2.1 Model Syntax

```

1 model2_syntax <- '
2   f1 =~ x1 + x2 + x3
3   f2 =~ x4 + x5 + x6
4 '
5
6 model2_fit <- sem(model2_syntax, data = my_data,
7                   estimator = "ML")

```

Listing A.2: Complete Model 2 specification

A.2.2 Population Parameters

Table A.2: Population parameters for Model 2 (Two-Factor CFA).

Parameter	Description	Value
λ_1	Loading of x_1 on f_1	0.80
λ_2	Loading of x_2 on f_1	0.70
λ_3	Loading of x_3 on f_1	0.90
λ_4	Loading of x_4 on f_2	0.70
λ_5	Loading of x_5 on f_2	0.80
λ_6	Loading of x_6 on f_2	0.60
ϕ_{11}	Variance of f_1	1.00
ϕ_{22}	Variance of f_2	1.00
ϕ_{12}	Covariance of f_1, f_2	0.50
δ_1 – δ_6	Residual variances	computed

A.2.3 Identification

- $p = 6$ (observed variables)
- $t = 13$ (free parameters: 4 loadings, 5 residual variances, 1 factor correlation, 3 fixed for identification)
- Identification constraints: λ_1 fixed to 0.8, λ_4 fixed to 0.7, $\phi_{11} = \phi_{22} = 1.0$
- Degrees of freedom: $df = 6(7)/2 - 13 = 21 - 13 = 8$

A.3 Model 3: Mediation Model

A.3.1 Model Syntax

```

1 model3_syntax <- '
2   m ~ a*x
3   y ~ c*x + b*m
4   '
5
6 model3_fit <- sem(model3_syntax, data = my_data,
7                   estimator = "ML")

```

Listing A.3: Complete Model 3 specification

A.3.2 Population Parameters

Table A.3: Population parameters for Model 3 (Mediation).

Parameter	Description	Value
a	Path $x \rightarrow m$	0.40
b	Path $m \rightarrow y$	0.30
c	Direct path $x \rightarrow y$	0.20
σ_x^2	Variance of x	1.00
ψ_m	Residual variance of m	0.84
ψ_y	Residual variance of y	computed

A.3.3 Identification

- $p = 3$ (observed variables: x, m, y)
- $t = 5$ (free parameters: $a, b, c, \psi_m, \psi_y; \sigma_x^2$ identified via just-identification)
- Degrees of freedom: $df = 3(4)/2 - 5 = 6 - 5 = 1$

- Indirect effect: $ab = 0.40 \times 0.30 = 0.12$
- Total effect: $c + ab = 0.20 + 0.12 = 0.32$

Appendix B

R Code Framework

This appendix provides representative code snippets from the Monte Carlo simulation implementation. Complete code is available in the accompanying files.

B.1 Main Simulation Loop

```
1 library(lavaan)
2 library(semTools)
3 library(MASS)
4
5 # Simulation parameters
6 n_rep      <- 1000
7 N_values   <- c(20, 30, 40, 50, 60, 80, 100, 150,
8               200, 300, 500, 1000)
9 alpha      <- 0.05
10 B_boot     <- 999
11
12 # Population covariance matrices
13 Sigma1 <- ... # Model 1: 3x3 matrix
14 Sigma2 <- ... # Model 2: 6x6 matrix
15 Sigma3 <- ... # Model 3: 3x3 matrix
16
17 # Model syntax
18 syntax1 <- ' f1 =~ x1 + x2 + x3 '
19 syntax2 <- ' f1 =~ x1 + x2 + x3\n f2 =~ x4 + x5 + x6 '
20 syntax3 <- ' m ~ a*x\n y ~ c*x + b*m '
21
22 # Storage
23 results <- array(NA, dim = c(3, 12, 7, n_rep))
24
25 # Main loop
26 for (m in 1:3) {                                # models
27   for (n in 1:12) {                              # sample sizes
```

```

28   cond_idx <- (m - 1) * 12 + n
29   N <- N_values[n]
30   Sigma <- switch(m, Sigma1, Sigma2, Sigma3)
31   syntax <- switch(m, syntax1, syntax2, syntax3)
32
33   for (i in 1:n_rep) {           # replications
34     seed <- i + cond_idx * 1000
35     result <- tryCatch({
36       set.seed(seed)
37       data <- mvrnorm(N, mu = rep(0, ncol(Sigma)),
38                       Sigma = Sigma)
39
40       # Fit models
41       fit_ml <- sem(syntax, data, estimator = "ML")
42       fit_mlr <- sem(syntax, data, estimator = "MLR")
43       fit_mlm <- sem(syntax, data, estimator = "MLM")
44
45       # Extract statistics and apply corrections
46       T_ML <- fitMeasures(fit_ml, "chisq")
47       T_SB <- ... # scaledTestStatistic()
48       T_YB <- ... # MLM mean-adjusted
49       T_Bart <- ... # chisqSmallN()
50       T_Swain <- ... # chisqSmallN(swain = TRUE)
51       T_F <- ... # F-correction
52       T_BS <- ... # bollenStineBootstrap()
53
54       # Record rejection decisions
55       c(pchisq(T_ML, df, lower.tail = FALSE) < alpha,
56         pchisq(T_SB, df, lower.tail = FALSE) < alpha,
57         ...)
58     }, error = function(e) rep(NA, 7))
59
60     results[m, n, , i] <- result
61   }
62 }
63 }

```

Listing B.1: Main simulation loop structure (pseudocode)

B.2 Wrapper Function: safe_chisqSmallN

```

1 safe_chisqSmallN <- function(fit, alpha = 0.05, swain = FALSE) {
2   tryCatch({
3     stat <- fitMeasures(fit, "chisq")
4     df <- fitMeasures(fit, "df")
5     n <- nobs(fit)

```

```

6
7   # semTools small sample chi-square
8   cs_result <- chisqSmallN(fit, swyn = swain)
9   T_corrected <- cs_result$chisq
10  p_value <- pchisq(T_corrected, df, lower.tail = FALSE)
11
12  list(reject = p_value < alpha,
13       p_value = p_value,
14       statistic = T_corrected,
15       status = "OK")
16 }, error = function(e) {
17   list(reject = NA, p_value = NA,
18        statistic = NA, status = paste("ERROR:", e$message))
19 })
20 }

```

Listing B.2: Wrapper function for Bartlett/Swain correction

B.3 Wrapper Function: safe__bollenstine

```

1 safe_bollenstine <- function(fit, B = 999, alpha = 0.05) {
2   tryCatch({
3     bs_result <- bollenStineBootstrap(fit, B = B)
4     p_value <- bs_result$pvalue
5
6     list(reject = p_value < alpha,
7          p_value = p_value,
8          statistic = NA,
9          status = "OK")
10  }, error = function(e) {
11    list(reject = NA, p_value = NA,
12         statistic = NA, status = paste("ERROR:", e$message))
13  })
14 }

```

Listing B.3: Wrapper function for Bollen-Stine bootstrap

B.4 Wrapper Function: safe__reject

```

1 safe_reject <- function(fit, alpha = 0.05) {
2   tryCatch({
3     T_ML <- fitMeasures(fit, "chisq")
4     df1 <- fitMeasures(fit, "df")
5     n <- nobs(fit)

```

```

6      t      <- length(fitMeasures(fit))  # or parameterCount()
7
8      # F-correction parameters
9      p      <- ncol(fit@Data@X)
10     a      <- 1 - (2*t + 2*p + 3) / (2*n - 2*t - 2*p - 1)
11     df2     <- (2*df1 + 4) / max(a^2, 0.001)
12
13     # F-statistic
14     F_stat  <- (T_ML / df1) / max(a, 0.001)
15     p_value <- pf(F_stat, df1, df2, lower.tail = FALSE)
16
17     list(reject = p_value < alpha,
18          p_value = p_value,
19          statistic = F_stat,
20          status = "OK")
21 }, error = function(e) {
22     list(reject = NA, p_value = NA,
23          statistic = NA, status = paste("ERROR:", e$message))
24 })
25 }

```

Listing B.4: Wrapper function for F-corrected LRT

Bibliography

- [1] Bartlett, M. S. (1937). Properties of sufficiency and statistical tests. *Proceedings of the Royal Society of London. Series A*, 160(901), 268–282.
- [2] Bollen, K. A., & Stine, R. A. (1992). Bootstrapping goodness-of-fit measures in structural equation models. *Sociological Methodology*, 22, 111–135.
- [3] Bollen, K. A., & Stine, R. A. (1993). Bootstrap inference in SEM. In K. A. Bollen & J. S. Long (Eds.), *Testing Structural Equation Models* (pp. 111–135). Sage.
- [4] Bradley, J. V. (1978). Robustness? *British Journal of Mathematical and Statistical Psychology*, 31(2), 144–152.
- [5] Browne, M. W. (1982). Covariance structures. In D. M. Hawkins (Ed.), *Topics in Applied Multivariate Analysis* (pp. 72–141). Cambridge University Press.
- [6] Fouladi, R. T. (2000). Performance of modified test statistics in covariance structure analysis under nonnormality and in large samples. *Structural Equation Modeling*, 7(2), 230–254.
- [7] Lawley, D. N. (1956). A general method for approximating to the distribution of likelihood ratio criteria. *Biometrika*, 43(1/2), 295–303.
- [8] Rosseel, Y. (2012). *lavaan*: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2), 1–36.
- [9] Satorra, A., & Bentler, P. M. (1988). Scaling corrections for chi-square statistics in covariance structure analysis. *Proceedings of the American Statistical Association*, 308–313.
- [10] Satorra, A., & Bentler, P. M. (1994). Corrections to test statistics and standard errors in covariance structure analysis. In A. von Eye & C. C. Clogg (Eds.), *Latent Variables Analysis: Applications for Developmental Research* (pp. 399–419). Sage.
- [11] Satorra, A., & Bentler, P. M. (2001). A scaled difference chi-square test statistic for moment structure analysis. *Psychometrika*, 66(4), 507–514.

- [12] Shapiro, A. (1984). A note on the consistency of the maximum likelihood estimator in the presence of incidental parameters. *Journal of the American Statistical Association*, 79(385), 134–138.
- [13] Swain, A. J. (1975). A class of factor analysis estimation procedures with common asymptotic sampling properties. *Psychometrika*, 40(3), 315–335.
- [14] Yuan, K.-H., & Bentler, P. M. (1997). Mean and covariance structure analysis: Theoretical and practical improvements. *Journal of the American Statistical Association*, 92(438), 767–774.
- [15] Yuan, K.-H., & Bentler, P. M. (1998). Normal theory based test statistics in structural equation modelling. *British Journal of Mathematical and Statistical Psychology*, 51(2), 289–309.
- [16] Wilks, S. S. (1938). The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics*, 9(1), 60–62.